such as a crop failure or a flooding, resulted from natural causes that could be identified through observation and reason. Further, he believed that if events could be rationally explained and understood, they could also be predicted. This approach to life marked a significant shift in human thinking.

The nature of the Universe

Another fundamental leap made by Thales was to question the nature of the Universe and what constitutes matter. He believed that there must be a single substance from which everything in the cosmos derives. Thales reasoned that water was the most likely candidate because it was vital to all forms of life, was capable of motion, and existed in different forms (solid, liquid, gas). He concluded that the substance in question must be water.

He also questioned the position of the Earth in the Universe and made the cosmological proposition that it must exist as a flat disk floating on water. His theory rested on the logic that every land mass is surrounded by water (the sea) and that animals, plants, and the Earth itself all absorb water. The significance of both theorems lies not in Thales' conclusions, which were incorrect, but in his desire to question the world and explain it rationally.

Applying science to life

Thales is praised for his grasp of mathematics and geometry. In an example cited by Herodotus, he calculated the height of an Egyptian pyramid by measuring the length of his own shadow and that of the pyramid at the same time of day, then applying the ratio he found between his shadow length and his height to that of the pyramid's shadow.

> **"Day** became **night**, and this change... **Thales** the Milesian had **foretold."**
>
> **Herodotus**, 5th century BCE